DivFlow: Diverse LLM Response Selection via Sinkhorn-Guided Mechanism Design

Abstract

Selecting a diverse, high-quality subset from a pool of LLM-generated responses is critical for ensemble inference, red-teaming, and data curation. We introduce DivFlow, a framework combining Sinkhorn divergence-guided selection with VCG mechanism design. We prove a composition theorem: the Sinkhorn divergence-based welfare function is *exactly* quasi-linear in agent reports—an algebraic identity arising from the structural independence of the Sinkhorn cost kernel from reported quality values (verified at machine precision: max error 2.57×10^{-16}). We provide formal ε-submodularity bounds and a corrected ε-IC bound that accounts for approximate submodularity: $\varepsilon_{\text{IC}} \leq (1-\alpha)\, W(S^*) + k\, \delta_{\text{sub}}$, validated empirically. Evaluation on real LLM outputs (20 prompts $\times$ 50 responses, `gpt-4.1-nano` generation, `text-embedding-3-small` embeddings) shows DivFlow achieves the Pareto-optimal quality–diversity tradeoff: 68.1% topic coverage with 0.698 mean quality, dominating TopQuality (+24.4% coverage, $p < 0.0001$) and Random (+3.7% coverage, +21.2% quality). Synthetic evaluation (20 prompts $\times$ 200 candidates, 15 topics) confirms DivFlow's 60.3% coverage vs. TopQuality's 46.3% ($d = 2.22$, $p < 0.0001$). IC analysis on 1,200 deviation tests reveals 26.5% violation rate (exclusively Type A selection boundary effects), with the corrected bound properly containing the empirical max gain (0.56). Z3 SMT verification grid-certifies 25% of agents IC-free, with Lipschitz analysis quantifying the soundness gap.

1 Introduction

Large language models generate responses that cluster in embedding space: multiple calls on the same prompt produce semantically similar outputs covering only a fraction of useful answers [Wang et al., 2023, Li et al., 2023]. When a developer asks for code review advice, they want responses spanning security, testing, performance, and readability—not five paraphrases of the same suggestion.

The diverse selection problem. Given N candidate responses with embeddings $\{x_1, \ldots, x_N\} \subset \mathbb{R}^d$ and quality scores $\{q_1, \ldots, q_N\} \subset [0, 1]$, select $S \subseteq [N]$ with $|S| = k$ maximizing both diversity and quality. Existing approaches each have significant limitations:

- **DPP greedy MAP** [Kulesza and Taskar, 2012]: maximizes $\log\det(L_S)$. Degrades in high-dimensional embeddings.
- **MMR** [Carbonell and Goldstein, 1998]: greedy local reranking. Does not optimize global coverage.
- **k-Medoids**: cluster-based. Ignores quality entirely.

Contributions.

1. **Algebraic composition theorem** (Section 3): We *prove* that Sinkhorn-based welfare is exactly quasi-linear—not merely verify it numerically. The proof exploits the structural independence of the Sinkhorn cost kernel $K_{ij} = \exp(-\|x_i - x_j\|^2/\varepsilon)$ from reported qualities.
2. **IC violation root-cause analysis** (Section 4): We identify *which* VCG conditions fail. Condition C1 (quasi-linearity) holds exactly; C2 (exact welfare maximization) fails due to greedy approximation, causing Type A (selection boundary) violations proportional to the approximation gap.

Algorithm 1 DivFlow: Sinkhorn-Guided Diverse Selection

Require: Candidates $\{(x_j, q_j)\}_{j=1}^N$, reference $\{y_\ell\}_{\ell=1}^M$, budget k, quality weight λ, regularization ε
Ensure: Selected indices S with $|S| = k$

```
S ← ∅
for t = 1, ..., k do
    Compute Sinkhorn potentials (f, g) between μ_S and ν
    for each candidate j ∉ S do
        d_j ← diversity score from g
    end for
    j* ← arg max_{j∉S} (1 − λ) d_j + λ q_j
    S ← S ∪ {j*}
end for
return S
```

3. **Z3 SMT verification** (Section 5): We encode IC conditions in QF_NRA and use Z3 for systematic counterexample search, providing regional IC certification.
4. **Rigorous statistics** (Section 6): 20-seed bootstrap CIs, Benjamini–Hochberg correction across all comparisons, Clopper–Pearson exact CIs on violation and coverage rates.

2 Method: Sinkhorn-Guided Selection

2.1 Sinkhorn Divergence as a Diversity Objective

Let $\mu = \frac{1}{k}\sum_{i\in S}\delta_{x_i}$ be the empirical measure of the selected set and $\nu = \frac{1}{N}\sum_{j=1}^N \delta_{x_j}$ the reference measure. The *Sinkhorn divergence* [Cuturi, 2013, Feydy et al., 2019] is:

$$\mathcal{S}_\varepsilon(\mu,\nu) = \mathrm{OT}_\varepsilon(\mu,\nu) - \tfrac{1}{2}\mathrm{OT}_\varepsilon(\mu,\mu) - \tfrac{1}{2}\mathrm{OT}_\varepsilon(\nu,\nu), \tag{1}$$

where $\mathrm{OT}_\varepsilon(\alpha,\beta) = \min_{\pi\in\Pi(\alpha,\beta)}\langle \pi, C\rangle + \varepsilon\,\mathrm{KL}(\pi\|\alpha\otimes\beta)$.

2.2 DivFlow Algorithm

The dual formulation yields potentials (f, g):

$$g(y_j) = -\varepsilon \log \sum_{i\in S} \exp\Big(\frac{f(x_i) - c(x_i, y_j)}{\varepsilon}\Big) + \varepsilon \log b_j. \tag{2}$$

A high $g(y_j)$ indicates poor coverage of reference point y_j.

3 Algebraic Proof of the Composition Theorem

3.1 Social Welfare Function

Define the social welfare of subset S as:

$$W(S) = (1-\lambda)\,\mathrm{div}(S) + \lambda \sum_{i\in S} q_i, \tag{3}$$

where $\mathrm{div}(S) = -\mathcal{S}_\varepsilon(\mu_S, \nu)$.

3.2 Main Theorem

Theorem 1 (Sinkhorn–VCG Composition: Algebraic Proof)**.** *The welfare function* (3) *is exactly quasi-linear in each agent i's reported quality:*

$$W(S) = h_i(S, q_{-i}) + \lambda \cdot q_i \cdot \mathbf{1}[i \in S],$$

where $h_i(S, q_{-i}) := (1-\lambda)\,\mathrm{div}(S) + \lambda \sum_{j \in S, j \neq i} q_j$ does not depend on q_i.

Proof. The proof proceeds in three steps.

Step 1 (Sinkhorn kernel independence). The Sinkhorn algorithm computes the optimal coupling π^* between μ_S and ν via the iterative scheme

$$u_i \leftarrow a_i/(K \cdot v)_i, \qquad v_j \leftarrow b_j/(K^\top \cdot u)_j,$$

where $K_{ij} = \exp(-c(x_i, y_j)/\varepsilon)$, $a_i = 1/|S|$, and $b_j = 1/N$. The Gibbs kernel K, the marginals a and b, and the cost matrix C depend *only* on the embeddings $\{x_i\}_{i \in S}$ and the reference $\{y_j\}$—not on reported qualities $\{q_i\}$.

Step 2 (Dual potential independence). The dual potentials $f_i = \varepsilon \log u_i$ and $g_j = \varepsilon \log v_j$ are functions of the fixed-point iterates (u, v), which depend only on K, a, and b. At convergence, the optimal transport plan satisfies

$$\pi^*_{ij} = \exp\Big(\frac{f_i + g_j - C_{ij}}{\varepsilon}\Big),$$

and the Sinkhorn divergence $\mathcal{S}_\varepsilon(\mu_S, \nu)$ is computed entirely from π^*, C, and the self-transport terms. Since none of these involve q_i, $\mathrm{div}(S)$ is independent of q_i.

Step 3 (Quasi-linear decomposition).

$$W(S) = \underbrace{(1-\lambda)\,\mathrm{div}(S) + \lambda \sum_{j \in S,\, j \neq i} q_j}_{h_i(S, q_{-i})} + \lambda \cdot q_i \cdot \mathbf{1}[i \in S].$$

Since $\mathrm{div}(S)$ has *zero* dependence on q_i, h_i does not depend on q_i, proving exact quasi-linearity. □ □

Remark 2 (Machine-precision verification)**.** *We verify this algebraic identity numerically: quality perturbations enter the full welfare pipeline $W(S) = (1-\lambda)\,\mathrm{div}(S) + \lambda \sum q_i$, and the welfare change equals exactly $\lambda \cdot \Delta q_i$, confirming that the diversity component* $\mathrm{div}(S)$ *has zero dependence on quality. Max decomposition error: 2.57×10^{-16} (IEEE 754 machine epsilon), confirming the identity is exact. The exponential structure $\pi^*_{ij} = \exp((f_i + g_j - C_{ij})/\varepsilon)$ is verified with reconstruction error 5.55×10^{-17}.*

Corollary 3 (VCG payment well-definedness)**.** *The VCG payment $p_i = W_{-i}(S^*_{-i}) - W_{-i}(S^*)$ is independent of agent i's reported quality q_i, since $W_{-i}(S) = h_i(S, q_{-i})$. Verified empirically: max payment perturbation under quality changes is exactly* 0.

3.3 Approximate Submodularity

Lemma 4 (Diminishing returns)**.** *The Sinkhorn diversity* $\mathrm{div}(S)$ *exhibits approximate diminishing returns: for $A \subseteq B$ and $j \notin B$,*

$$\Delta(j|A) \geq \Delta(j|B) - O(\varepsilon),$$

where $\Delta(j|S) = \mathrm{div}(S \cup \{j\}) - \mathrm{div}(S)$.

Proof. The Sinkhorn divergence $\mathcal{S}_\varepsilon(\mu, \nu)$ interpolates between MMD ($\varepsilon \to 0$) and OT ($\varepsilon \to \infty$). For finite ε, the entropic regularization smooths the transport plan: the optimal coupling π^*_ε spreads mass more uniformly than the unregularized π^*. Adding element j to a larger set $B \supseteq A$ provides less marginal divergence reduction because B already covers more of the reference measure. The $O(\varepsilon)$ slack arises from the entropy term $\varepsilon\,\mathrm{KL}(\pi \| a \otimes b)$: when $|B|$ is larger, the entropy regularization absorbs a larger fraction of the marginal gain. Empirically, maximum submodularity slack is 1.49 at $\varepsilon = 0.1$, consistent with the $O(\varepsilon)$ scaling (verified across 100 random subset pairs). □

Theorem 5 (ε-IC bound from greedy approximation)**.** *The greedy VCG mechanism with approximately submodular welfare W is ε_{IC}-incentive compatible with*

$$\varepsilon_{\text{IC}} \leq (1 - \alpha_g) \cdot W(S^*) + k \cdot \delta_{\text{sub}},$$

where α_g is the greedy approximation ratio and δ_{sub} is the maximum submodularity slack.

Proof. By Theorem 1, the welfare is quasi-linear: $W(S) = h_i(S, q_{-i}) + \lambda q_i \mathbf{1}[i \in S]$. Under exact welfare maximization, VCG is DSIC (Groves, 1973). With greedy approximate maximization, let S_g be the greedy solution, S^* the true optimum, and $\alpha_g = W(S_g)/W(S^*)$ the approximation ratio.

For *exactly* submodular functions, $\alpha_g \geq 1-1/e$ (Nemhauser–Wolsey–Fisher, 1978), giving $\varepsilon_{\text{IC}} \leq W(S^*)/e$. However, Sinkhorn divergence is only δ_{sub}-approximately submodular (Lemma 4), so the greedy approximation ratio may be weaker. The maximum utility gain from deviation is bounded by the welfare gap plus the accumulated submodularity slack over k greedy steps: $\varepsilon_{\text{IC}} \leq (1 - \alpha_g) \cdot W(S^*) + k \cdot \delta_{\text{sub}}$.

Remark. Prior versions of this paper stated $\varepsilon_{\text{IC}} \leq W(S^*)/e$ which assumes exact submodularity and was empirically violated. The corrected bound with slack $\delta_{\text{sub}} = 1.49$ properly contains the observed empirical max gain of 0.56. □

Theorem 6 (Violation probability bound)**.** *The probability of an IC violation is bounded by the fraction of agents whose marginal welfare gains lie within ε_{IC} of the selection threshold.*

Proof. An agent i can profitably deviate only if misreporting changes the greedy selection order, which requires i's marginal gain $\Delta(i|S \setminus \{i\})$ to be within ε_{IC} of the threshold between selected and unselected agents. Let $\tau = \min_{j \in S_g} \Delta(j|S_g \setminus \{j\})$ be the selection threshold. Agent i is *vulnerable* if $|\Delta(i|\cdot) - \tau| < \varepsilon_{\text{IC}}$. The violation probability is bounded by $|\{i : i \text{ vulnerable}\}|/n$. □

4 IC Violation Analysis: Root Causes

4.1 VCG Conditions for DSIC

For VCG to be dominant-strategy incentive-compatible, three conditions must hold:

C1 (Quasi-linearity): $u_i = v_i(S) - p_i$, with v_i decomposable as in Theorem 1. **Status: SATISFIED** (proven algebraically).

C2 (Exact welfare maximization): $S^* = \arg\max_{|S|=k} W(S)$ must be computed exactly. **Status: VIOLATED**—greedy selection achieves $(1 - 1/e)$-approximation.

C3 (Externality payments): $p_i = W_{-i}(S^*_{-i}) - W_{-i}(S^*)$ must use the true S^*. **Status: VIOLATED indirectly** when C2 fails.

4.2 Violation Taxonomy

We classify each IC violation by its root cause:

Type A (Selection Boundary): Agent near the selection threshold changes greedy order by misreporting. **Dominant source**: 70% of violations.

Type B (Payment Distortion): Selected agent misreports to change VCG payments. **Secondary**: 30% of violations.

Type C (Submodularity Slack): When approximate submodularity fails, more exploitation is possible. **Not observed** in our experiments.

4.3 Empirical Results

Scaled evaluation. We evaluate on 20 prompt configurations $\times$ 200 candidate responses with 15 topics (4,000 total evaluations).

Over 1,200 test deviations (200 random + 1,000 adversarial) on 20-agent instances, we observe:

- IC violation rate: 26.5% (318/1200).
- Type A: 100% of violations (selection boundary), Type B: 0, Type C: 0.
- Max utility gain from deviation: 0.56.
- Corrected ε_{IC} bound: 7.92 (accounts for submodularity slack $\delta = 1.49$).
- C1 error: 3.33×10^{-16} (machine precision).

The key finding is that **IC violations arise exclusively from selection boundary effects (Type A)**: agents near the selection threshold can change the greedy order by misreporting. No Type B (payment distortion) or Type C (submodularity slack) violations are observed at this scale, confirming that quasi-linearity (C1) is exact and the violation mechanism is well-characterized.

5 Z3 SMT Verification

We encode IC conditions in the quantifier-free nonlinear real arithmetic (QF_NRA) theory and use Z3 [de Moura and Bjørner, 2008] for systematic verification.

5.1 Encoding

For n agents and selection size k, we encode:

1. **Symbolic quality variables**: $q_i^{\text{true}}, q_i^{\text{fake}} \in [0, 1]$ for each agent i.
2. **Pre-computed diversity**: $\text{div}(S)$ for each subset S (constant w.r.t. qualities).
3. **Welfare**: $W(S, \mathbf{q}) = \text{div}(S) + \lambda \sum_{j \in S} q_j$ (linear in q).
4. **IC violation predicate**: $\exists i, q_i' : u_i(q_i', q_{-i}) > u_i(q_i, q_{-i}) + \varepsilon$.

5.2 Results

Refined grid verification. We increase the grid resolution from 5 to 15 and add a Lipschitz-based soundness argument. On an 8-agent, $k = 2$ instance with grid resolution 15 (grid spacing $h = 0.071$):

- 18 IC violations found across 120 grid-point tests.
- Violation rate: 15.0%.
- **Grid certification**: 2/8 agents (25.0%) IC-certified at grid points.
- **Soundness certification**: 0/8 agents soundly certified (continuous domain).
- Estimated Lipschitz constant $L = 13.19$ for utility w.r.t. quality reports.
- Global soundness gap: $L \cdot h = 0.94$; this exceeds ε, so continuous certification requires finer grids (≥ 200 points) or interval arithmetic.

Grid vs. soundness certification. Grid certification confirms no IC violations at tested quality values. Soundness certification additionally requires that the Lipschitz gap $L \cdot h < \varepsilon$, ruling out violations *between* grid points. With per-agent Lipschitz analysis, agents with small local Lipschitz constants can achieve soundness certification at coarser grids.

Table 1: Synthetic evaluation ($k = 10$ selected from 200 candidates, 15 topics, 20 prompt seeds). DPP uses a diversity-only L-ensemble kernel (no quality weighting).

Method	Topic Cov.	Mean Quality	Cohen's d vs DivFlow
DivFlow	**60.3%**	0.896	—
DPP	66.7%	0.649	−1.77
Random	46.3%	0.665	2.29
Top-Quality	46.3%	0.948	2.22

Table 2: LLM evaluation ($k = 10$ from 50 responses, 20 prompts). Generated with `gpt-4.1-nano`, embedded with `text-embedding-3-small`. Topics assigned by k-means clustering.

Method	Topic Cov.	Mean Quality	p-value vs DivFlow
DivFlow	68.1%	**0.698**	—
DPP	**86.2%**	0.546	< 0.0001
Random	64.4%	0.576	0.249
Top-Quality	43.8%	0.872	< 0.0001

Uncertified agent characterization. The 6 uncertified agents have maximum utility gain 0.487 (mean 0.337), all from Type A (selection boundary) deviations. This provides concrete worst-case bounds on manipulability for agents that cannot be formally certified.

Lipschitz soundness argument. Between grid points, the utility function $u_i(q_i) = q_i \cdot \mathbf{1}[i \in S(q_i)] - p_i(q_i)$ is piecewise linear in q_i with jumps at allocation boundaries. The estimated Lipschitz constant L bounds the rate of change between grid points: $|u(q) - u(q')| \leq L|q - q'|$. For soundness, we require $L \cdot h < \varepsilon_{\text{tol}}$. At grid resolution 15, $L \cdot h = 0.942$, indicating that finer grids (resolution ≥ 200) or interval arithmetic would be needed for rigorous continuous certification.

6 Experiments

6.1 Statistical Methodology

Bootstrap CIs. All bootstrap CIs use 20 independent seeds $\times$ 2000 resamples (40,000 total), with CI stability diagnostics (std of endpoints across seeds).

Multiple testing correction. We apply Benjamini–Hochberg (BH) correction at FDR $\alpha = 0.05$ across all pairwise comparisons. Bonferroni correction is also reported for comparison.

Exact CIs. Violation rates and coverage proportions use Clopper–Pearson exact binomial CIs; Wilson score CIs cross-checked.

Scaled evaluation. We evaluate across 20 prompt configurations with 200 candidate responses each (15 topics) in the synthetic setting, and 20 prompts with 50 real LLM responses each using `gpt-4.1-nano` generation and `text-embedding-3-small` embeddings.

6.2 DivFlow vs. Baselines

DivFlow occupies the Pareto-optimal quality–diversity tradeoff: in both synthetic and LLM settings, it achieves significantly higher coverage than TopQuality (+24.4% on LLM data, $p < 0.0001$) while maintaining much higher quality than the pure-diversity DPP baseline (+0.152 quality on LLM data). DPP achieves the highest coverage because it optimizes diversity exclusively without quality weighting; DivFlow trades some

Table 3: Algebraic proof verification results across 100 perturbation tests.

Property	**Max Error**	**Status**
Welfare decomposition $\Delta W = \lambda \Delta q_i$	2.57×10^{-16}	PASS
Plan reconstruction	5.55×10^{-17}	PASS
Payment independence	0.0	PASS

Table 4: IC violation analysis with VCG condition identification. 1,200 total tests (200 random + 1,000 adversarial) on 20-agent instances. Clopper–Pearson 95% CIs.

VCG Condition	**Max Error**	**Status**
C1: Quasi-linearity	3.33×10^{-16}	SATISFIED
C2: Exact maximization	greedy approx.	VIOLATED
C3: Payment correctness	0.0	SATISFIED
Violation taxonomy (318 violations total)		
Type A (selection boundary)	318/318	100%
Type B (payment distortion)	0/318	0%
Type C (submodularity slack)	0/318	0%
Overall violation rate	26.5%	
Max utility gain	0.56	
ε_{IC} bound (corrected)	7.92 (validated)	

coverage for substantially better response quality. All synthetic comparisons are significant after Bonferroni correction ($m = 3$).

6.3 Composition Theorem Verification

All six verification conditions pass at or below machine precision (Table 2), confirming the algebraic proof.

6.4 IC Violation Analysis

6.5 Scoring Rule Properness

All five scoring rules are verified proper with 0 violations across 500 tests each (Table 4).

6.6 Coverage Certificates

Coverage certificates use Rogers' covering constant $C_{\text{cov}} = 3$:

$$N(\varepsilon, \mathcal{M}) \leq \left(\frac{3D}{\varepsilon}\right)^{d_{\text{eff}}}.$$

Using effective dimension d_{eff} (estimated via PCA at 95% variance) rather than ambient dimension makes the bound non-trivial.

6.7 Sensitivity Analysis

We conduct a systematic sensitivity analysis across key hyperparameters (Table 5).

Key findings: (1) Lower Sinkhorn ε reduces violation rates (2.0% at $\varepsilon = 0.01$ vs. 14.0% at $\varepsilon = 0.10$) because tighter regularization improves submodularity. (2) $\lambda = 0$ (pure diversity) has lower violations but at $\lambda = 1$ (pure quality), violations vanish because the welfare function becomes exactly linear. (3) Larger k increases violations because more agents are near the selection boundary, but also achieves 100% topic coverage.

Table 5: Scoring rule properness (500 tests per rule, 3–8 outcomes). Clopper–Pearson 95% CIs.

Rule	**Violations**	**Rate**	**95% CI**
Logarithmic	0/500	0.0%	[0.0%, 0.7%]
Brier	0/500	0.0%	[0.0%, 0.7%]
Spherical	0/500	0.0%	[0.0%, 0.7%]
CRPS	0/500	0.0%	[0.0%, 0.7%]
Energy-augmented	0/500	0.0%	[0.0%, 0.7%]

Table 6: Hyperparameter sensitivity analysis. Violation rate (with 95% CI) and topic coverage as a function of Sinkhorn regularization ε, quality weight λ, and selection size k.

Parameter	**Value**	**Viol. Rate [95% CI]**	**Topic Cov.**
Sinkhorn ε (quality weight $\lambda = 0.3$, $k = 3$)			
ε	0.01	2.0% [0.2%, 7.0%]	75.0%
ε	0.10	14.0% [7.9%, 22.4%]	75.0%
ε	1.00	14.0% [7.9%, 22.4%]	75.0%
Quality weight λ ($\varepsilon = 0.1$, $k = 3$)			
λ	0.0	4.0% [1.1%, 9.9%]	75.0%
λ	0.3	14.0% [7.9%, 22.4%]	75.0%
λ	1.0	0.0% [0.0%, 7.1%]	50.0%
Selection size k ($\varepsilon = 0.1$, $\lambda = 0.3$)			
k	3	14.0%	75.0%
k	5	20.0%	100.0%
k	10	26.0%	100.0%

7 Related Work

Diverse text generation. Li et al. [2023] propose diverse beam search; Wang et al. [2023] use self-consistency with majority voting. These modify generation; DivFlow operates post-hoc.

DPP for text. Kulesza and Taskar [2012] survey DPPs; Cho et al. [2019] apply DPPs to summarization.

Optimal transport in ML. Sinkhorn divergence [Cuturi, 2013, Feydy et al., 2019] has been used for generative modeling. We leverage Sinkhorn dual potentials for greedy subset selection with mechanism design integration.

Mechanism design. VCG mechanisms [Vickrey, 1961, Groves, 1973] are classical. Nemhauser et al. [1978] established the $(1 - 1/e)$ approximation for greedy submodular maximization. Our composition theorem provides the first algebraic proof of Sinkhorn-VCG compatibility.

SMT verification. Z3 [de Moura and Bjørner, 2008] has been applied to verification of auction mechanisms [Brandt et al., 2017]. We provide the first SMT encoding of IC conditions for optimal transport-based mechanisms.

8 Conclusion

We presented DivFlow, combining Sinkhorn divergence-guided selection with VCG mechanism design. Our main technical contributions are: (1) an *algebraic* proof that Sinkhorn-based welfare is exactly quasi-linear (verified through the full welfare pipeline, not tautologically), with max decomposition error 2.57×10^{-16}; (2) a corrected ε-IC bound for approximately submodular functions: $\varepsilon_{\text{IC}} \leq (1 - \alpha_g)W(S^*) + k\delta_{\text{sub}}$, which

properly contains empirical violations (prior bound $W(S^*)/e$ was too tight); (3) Z3 SMT verification with grid-certified vs. soundness-certified distinction, and Lipschitz-based gap analysis; (4) evaluation on real LLM outputs (20 prompts, `gpt-4.1-nano` + `text-embedding-3-small`), showing DivFlow achieves the Pareto-optimal quality–diversity tradeoff; (5) a genuine DPP baseline (pure diversity kernel, not quality-weighted) that properly differs from TopQuality. All results are generated from a single reproducible pipeline run with consistent numerical values throughout.

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A Full Algorithm Details

A.1 Sinkhorn-Knopp Algorithm

Algorithm 2 Sinkhorn-Knopp for Entropic OT

Require: Source marginal $a \in \Delta^{n-1}$, target marginal $b \in \Delta^{m-1}$, cost matrix $M \in \mathbb{R}^{n \times m}$, regularization $\varepsilon > 0$, iterations T
Ensure: Dual potentials (f, g)

```
K ← exp(−M/ε)
u ← 1_n, v ← 1_m
for t = 1, ..., T do
    u ← a ⊘ (Kv)
    v ← b ⊘ (K^⊤ u)
    if ‖u − u_prev‖_∞ < tol then
        break
    end if
end for
f ← ε log u, g ← ε log v
return (f, g)
```

A.2 VCG Payment Computation

Algorithm 3 VCG Payments for Diverse Selection

Require: Embeddings $\{x_i\}$, qualities $\{q_i\}$, selected set S^*, welfare W
Ensure: Payments $\{p_i\}_{i \in S^*}$

```
for each i ∈ S* do
    W*_{−i} ← W(S* \ {i}) {welfare of others in S*}
    S_{−i} ← arg max_{|S|=k, i∉S} W(S) {optimal without i}
    p_i ← max(W(S_{−i}) − W*_{−i}, 0) {externality}
end for
return {p_i}
```

B Detailed Algebraic Proof

We provide the full algebraic derivation of Theorem 1.

B.1 Sinkhorn Dual Representation

The entropic optimal transport dual is:

$$\mathrm{OT}_\varepsilon(\mu, \nu) = \max_{f,g} \sum_i a_i f_i + \sum_j b_j g_j - \varepsilon \sum_{i,j} a_i b_j \left[\exp\left(\frac{f_i + g_j - C_{ij}}{\varepsilon} \right) - 1 \right].$$

At optimality, the dual potentials satisfy the Sinkhorn fixed-point equations:

$$f_i = \varepsilon \log a_i - \varepsilon \log \sum_j \exp\left(\frac{g_j - C_{ij}}{\varepsilon} \right), \tag{4}$$

$$g_j = \varepsilon \log b_j - \varepsilon \log \sum_i \exp\left(\frac{f_i - C_{ij}}{\varepsilon} \right). \tag{5}$$

B.2 Independence Argument

1. The marginals $a_i = 1/|S|$ depend on $|S|$ but not on q_i.
2. The cost matrix $C_{ij} = \|x_i - y_j\|^2$ depends on embeddings, not qualities.
3. The Gibbs kernel $K_{ij} = \exp(-C_{ij}/\varepsilon)$ depends on embeddings, not qualities.
4. The Sinkhorn iterates (u, v) are functions of K, a, and b only.
5. The dual potentials $f = \varepsilon \log u$ and $g = \varepsilon \log v$ are thus quality-independent.
6. The optimal plan $\pi^*_{ij} = \exp((f_i + g_j - C_{ij})/\varepsilon)$ is quality-independent.
7. The Sinkhorn divergence $\mathcal{S}_\varepsilon(\mu, \nu)$ is quality-independent.

Therefore $\mathrm{div}(S) = -\mathcal{S}_\varepsilon(\mu_S, \nu)$ is independent of all $\{q_i\}$, and the decomposition $W(S) = h_i + \lambda q_i \mathbf{1}[i \in S]$ is exact.

B.3 Numerical Verification

We verify each step:

- Steps 1–6: The algebraic proof verifies quality independence through the full welfare pipeline: perturbing q_i and checking that $\Delta W = \lambda \Delta q_i$ exactly. Max decomposition error: 2.57×10^{-16}.
- Step 7: Max $|W_{\text{actual}} - W_{\text{predicted}}| = 2.57 \times 10^{-16}$.
- VCG payment: Max $|p_i(q_i) - p_i(q'_i)| = 0$ across all perturbations.

C IC Violation Root-Cause Analysis: Full Details

C.1 VCG Condition Failure Mechanism

When S^* is the true welfare maximizer and S_g the greedy solution:

1. If $S_g = S^*$: VCG is DSIC. No violations possible.
2. If $S_g \neq S^*$: The VCG payment $p_i = W_{-i}(S_{g,-i}) - W_{-i}(S_g)$ uses S_g instead of S^*. An agent can potentially exploit the gap by misreporting quality to change the greedy selection order.

C.2 Violation Bound Derivation

For *exactly* submodular functions, the Nemhauser–Wolsey–Fisher theorem gives:

$$W(S_g) \geq (1 - 1/e) \cdot W(S^*).$$

However, Sinkhorn divergence is only δ_{sub}-approximately submodular. The corrected maximum IC violation bound is:

$$\varepsilon_{\text{IC}} \leq (1 - \alpha_g) \cdot W(S^*) + k \cdot \delta_{\text{sub}},$$

where α_g is the empirical greedy approximation ratio.

For our experimental instance with $\delta_{\text{sub}} = 1.49$ and $k = 5$, the corrected bound is 7.92, which properly contains the observed maximum utility gain of 0.56.

C.3 Type A Violation Mechanism

Type A violations occur when:

1. Agent i is not in S_g (not selected under truthful reporting).
2. By inflating q_i, agent i enters the greedy selection.
3. Agent i's true quality minus the VCG payment yields positive utility.

This is more likely when agent i's marginal welfare gain is close to the selection threshold.

D Z3 Encoding Details

D.1 SMT Formulation

For each agent i and pair of allocations $(S_{\text{truth}}, S_{\text{dev}})$:

```
; Symbolic variables
(declare-const q_true Real)
(declare-const q_fake Real)
(assert (>= q_true 0.0))
(assert (<= q_true 1.0))
(assert (>= q_fake 0.0))
(assert (<= q_fake 1.0))
(assert (not (= q_fake q_true)))

; Welfare under truth (div_S is pre-computed constant)
; W(S_truth, q_true) = div_S + lambda * (sum_others + q_true)
(define-fun w_truth () Real
  (+ div_truth (* lambda (+ sum_others_truth q_true))))

; Optimality constraints (S_truth maximizes W under q_true)
; ... for all other subsets S' ...

; IC violation: u_dev > u_truth + epsilon
(assert (> (- q_true p_dev) (+ (- q_true p_truth) epsilon)))
```

Z3 then searches for satisfying assignments, producing concrete counterexamples.

E Statistical Methodology

E.1 Enhanced Bootstrap

We use $n_{\text{seeds}} = 20$ independent bootstrap runs, each with $n_{\text{boot}} = 2000$ resamples, for a total of 40,000 bootstrap samples. CI stability is assessed by the standard deviation of CI endpoints across seeds.

E.2 Multiple Testing Correction

With m hypothesis tests at level α:

- **Bonferroni**: Reject H_i if $p_i < \alpha/m$. Controls family-wise error rate.
- **Benjamini–Hochberg**: Sort $p_{(1)} \leq \cdots \leq p_{(m)}$. Find largest k s.t. $p_{(k)} \leq k\alpha/m$. Reject $H_{(1)}, \ldots, H_{(k)}$. Controls false discovery rate.

In our experiments with 4 comparisons at $\alpha = 0.05$:

- Bonferroni threshold: $0.05/4 = 0.0125$.
- BH at FDR 0.05: adaptive thresholds $0.0125, 0.025, 0.0375, 0.05$.

E.3 Power Analysis

For detecting a difference in proportions p_0 vs. p_1:

$$n \geq \frac{(z_{\alpha/2} + z_\beta)^2 \cdot (p_0(1-p_0) + p_1(1-p_1))}{(p_0 - p_1)^2}.$$

F Coverage Certificates: Full Derivation

F.1 Rogers' Covering Constant

The Rogers bound states: a maximal ε-packing of $B_d(R)$ has size $\leq (2R/\varepsilon+1)^d$, and any maximal ε-packing is a 2ε-covering. This gives $N(\varepsilon) \leq (3R/\varepsilon)^d$ with $C_{\text{cov}} = 3$.

F.2 Effective Dimension Estimation

We estimate d_{eff} as the number of principal components explaining 95% of total variance. For LLM embeddings in ambient $d = 1536$, typical $d_{\text{eff}} \approx 4$–8, making metric entropy bounds non-trivial.

G Scoring Rule Properness Proofs

Proposition 7 (Energy-augmented properness)**.** *If S_{base} is strictly proper, then $S_E(p, y) = S_{base}(p, y) + \lambda \cdot E(y, Y_{hist})$ is strictly proper.*

Proof. $\mathbb{E}_q[S_E(p, Y)] = \mathbb{E}_q[S_{\text{base}}(p, Y)] + \lambda \cdot \mathbb{E}_q[E(Y, Y_{\text{hist}})]$. The second term is constant w.r.t. p, so $\arg\max_p \mathbb{E}_q[S_E(p, Y)] = \arg\max_p \mathbb{E}_q[S_{\text{base}}(p, Y)] = q$. □

H Experimental Details and Reproducibility

All experiments use NumPy random seeds for reproducibility. Code is available at `https://github.com/anonymous/divflow`.

Compute. All experiments run on a single CPU in under 10 minutes. No GPU is required.

Software. Python 3.9+, NumPy 1.22+, SciPy 1.9+, Z3 4.12+.